

Step-by-Step Documentation: Running a Python Socket Client-Server Application

Step 1: Install Python

- Visit <https://www.python.org/downloads/>
- Download and install Python 3.7 or later.
- During installation, make sure to check the box that says "Add Python to PATH".
- After installation, verify it using Command Prompt or Terminal:
 - > python --version
 - OR
 - > py --version

Step 2: Prepare Your Project Files

- Create a folder named `project-folder` (or any name you like).
- Inside this folder, save two files:
 1. server.py
 2. client.py
- Ensure both files are in the same directory.

Step 3: server.py Code

This file acts as the server. It listens on a specified port and replies to client messages.
Paste this code inside `server.py`:

```
import socket

server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_socket.bind(('localhost', 65432))
server_socket.listen()

print("Server listening on port 65432...")
conn, addr = server_socket.accept()

with conn:
    print('Connected by', addr)
    while True:
```

```
data = conn.recv(1024)
if not data:
    print("Connection closed by client.")
    break
print("Received from client:", data.decode())
reply = "Server received: " + data.decode()
conn.sendall(reply.encode())
```

Step 4: client.py Code

This file acts as the client. It connects to the server and sends messages.
Paste this code inside `client.py`:

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client_socket.connect(('localhost', 65432))

while True:
    message = input("Enter your message (type 'exit' to quit): ")
    if message.lower() == 'exit':
        print("Closing connection.")
        break
    client_socket.sendall(message.encode())
    data = client_socket.recv(1024)
    print('Received from server:', data.decode())

client_socket.close()
```

Step 5: Running the Server

- Open a terminal or CMD.
- Navigate to your project folder:
> cd path_to_project_folder
- Start the server:
> python server.py

Step 6: Running the Client

- Open another terminal or CMD window.
- Navigate to the same project folder.
- Start the client:
 > python client.py

Step 7: Start Messaging

- In the client terminal, type messages.
- The server will print received messages and reply to them.
- To exit, type `exit` from the client.

♣ Troubleshooting

- If Python is not recognized, use `py` instead of `python`.
- If port 65432 is busy, change it in both files to another port.
- If files not found, make sure you're in the correct directory.